Predictive Multi-Criteria Allocation across DeFi Lending Protocols:
A Forecast-Driven Extension of the AI-Managed ERC-4626 Vault on the `fractal-defi` Substrate

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Abstract

We study dynamic capital allocation between Aave V3 and Compound V3 USDC supply markets on Ethereum mainnet and ask whether replacing a reactive EMA-smoothed rate observation with a 12-hour-ahead learned forecast yields a significant improvement in risk-adjusted return. The proposed allocator combines a dual-branch DA-BiGRU-CNN forecaster — decomposing each protocol's supply rate into a deterministic kink component and a learned residual — with a four-factor TOPSIS-style multi-criteria decision rule inherited from the author's prior ERC-4626 yield vault [18], implemented as a concrete subclass of a new `BaseLendingAllocationStrategy` abstraction contributed to the `fractal-defi` research substrate [11, 12]. We assembled an 18-month hourly panel (13,096 Aave V3 and 12,904 Compound V3 bars, Nov 2024 – Apr 2026); on the 12,895-hour overlap we empirically find a cross-protocol spread with median -0.232 pp, standard deviation 2.513 pp, and an Aave-pays-more share of 40.7%, and we observe a regime shift between 2025 Q3 and 2025 Q4 where this share jumps 39.9% → 56.3% — the first quarter on the panel where Aave systematically pays more. An initial sparse-sample analysis (37% Compound coverage) had suggested a 2025 Q1→Q2 shift; the 98.5% full-coverage panel revealed that finding to be a cluster-sampling artifact, a methodological lesson we report explicitly. Five strategies (buy-and-hold, APY-only greedy, MCDM-EMA, CIR-MCDM, predictive MCDM) will be backtested with a 10/4/4-month split and fifteen ablations. Pre-registered targets are out-of-sample $R^2 \in [0.15, 0.40]$, 55–65% directional accuracy, and Sharpe improvement 0.2–0.5 over EMA with bootstrap $p < 0.05$; the null hypothesis is entertained explicitly and is structurally falsifiable.

1 Introduction

1.1 Motivation

Decentralized lending protocols offer USDC supply-side yield that fluctuates with utilization, market demand, and protocol-specific governance parameters. Supply rates on Aave V3 and Compound V3 are not synchronized: they cross repeatedly over time, and [7] establish empirically that Compound borrowing rates *lead* Aave, with cointegration coefficients implying Aave adjusts at speed 0.607 toward the long-run Compound relationship. A capital provider who can anticipate these crossovers and rebalance ahead of them captures a yield differential that a reactive observer misses. On the 18-month Aave V3 / Compound V3 USDC panel assembled for this work (Section 5), cross-protocol APR spreads cross zero in 40.7% of hours, and a regime shift in 2025 Q4 (Aave-pays-more share 39.9% → 56.3%, median spread crossing zero from -0.218 pp to

*Code and reproducibility artifacts: https://github.com/SergeySolovyev/predictive-mcdm-defi. This work realizes the methodological transfer outlined in Section 6.4 of [19] and previewed at the Scientific Telegraph Young Researcher Conference, Central University, Moscow, 17 May 2026.

+0.183 pp) is the first sub-period on the panel where Aave becomes the persistently-dominant payer — the kind of conditional structure that a forecast-driven allocator is designed to exploit. Yet the production-grade DeFi yield aggregators surveyed in Section 2 all act on observed (or EMA-smoothed) rates, and the public research literature on `fractal-defi`-based backtesting [11, 20, 21] covers AMM liquidity provision and basis trading, but not the *lending* microstructure panel of the same substrate. This paper fills that gap.

The forecasting problem is structurally analogous to high-frequency mid-price prediction in limit-order books: noisy multivariate time series, regime-dependent dynamics, a weighted-error metric in which large moves matter disproportionately, and a discrete decision rule (rebalance or hold). The methodological transfer from the author's LOB-prediction work [19] to the present DeFi-rate setting is the central scientific claim. The transfer is not mechanical: a hard domain prior (the protocol-known piecewise-linear "kink" rate function) is available in DeFi and absent in LOBs, and the architecture exploits it via residual subtraction.

1.2 Contributions

1. **Predictive MCDM allocator.** A 12-hour-ahead forecast replaces the reactive EMA observation in the four-factor TOPSIS-style MCDM rule of [18], producing the first published forecast-driven multi-criteria USDC allocator across Aave V3 and Compound V3 on mainnet historicals.
2. **Dual-branch forecaster with kink subtraction.** A DA-BiGRU-CNN architecture predicts each protocol's utilization and rate residual separately, then reconstructs the rate via the protocol-known kink function. This is the lending-rate analogue of the dual-branch LOB predictor of [19], with the kink prior as a genuine architectural contribution beyond either prior work.
3. **Composite forecast loss.** A weighted combination of MSE, weighted-Pearson correlation, and a quantile-90 term aligns the training objective with allocation-relevant accuracy (large moves and the upper tail of the rate distribution).
4. **Open-source contributions to `fractal-defi`.** Two pull requests of independent utility: a Compound V3 lending history loader with a uniform `utilization` field exposed on both Aave and Compound `GlobalState` objects (Extra+1), and a `BaseLendingAllocationStrategy` abstraction generalizing the multi-protocol allocator pattern (Extra+2).
5. **Honest negative-result protocol.** The null hypothesis (forecast adds no value over EMA on this horizon) is operationalized, pre-registered, and entertained as a publishable outcome; the fifteen-row ablation matrix is structured to attribute any observed effect to forecast, criterion structure, or protocol-pair selection.
6. **Empirical regime-structure observation on Aave V3 / Compound V3 supply rates that itself constitutes a research artifact.** The 12,895-hour full-coverage panel (Section 5) documents a 2025 Q3→Q4 regime shift — the first quarter in which Aave systematically pays more than Compound — and a $\sim 10\times$ intra-panel volatility variation. A subsidiary methodological finding (Section 5.3): an initial 37%-coverage analysis suggested a 2025 Q1→Q2 shift that the full 98.5% panel attributes to cluster sampling, an instructive negative result on regime-narrative robustness in sparse DeFi panels.

1.3 Paper Outline

Section 2 reviews lending mechanics, MCDM in financial allocation, deep-learning rate forecasting, the bridge to LOB microstructure, and the recent "fair interest rates are impossible" counter-result [8]. Section 3 situates the work on the `fractal-defi` substrate and describes the new `BaseLendingAllocationStrategy` abstraction. Section 4 develops the dual-branch forecaster and decision rule. Section 5 and Section 6 cover the 18-month data window and hyperparameter calibration. Section 7 states the backtesting protocol; Section 8 hosts the headline and per-baseline tables (with numerical entries deferred to Week 3 execution). **??** interprets the findings and addresses the Halpern–Pass–Saraf objection. Section 9 closes with limitations and future work.

2 Background and Related Work

2.1 DeFi Lending Mechanics: Aave V3 and Compound V3

Both Aave V3 and Compound V3 (Comet) implement piecewise-linear *kink* rate models: borrow APY rises linearly with utilization $u \in [0, 1]$ at slope s_1 up to an "optimal" utilization u^*, then steeply at slope s_2. Supply APY is derived from borrow APY, utilization, and a protocol-fixed reserve factor. Despite identical functional form, the two protocols differ in parameter calibration, governance cadence, and oracle architecture; the result is short-horizon dispersion in observed USDC supply rates with periodic crossovers, as analyzed empirically in [3, 7]. The reserve-factor adjustment on either side is not learned from data: it is published on-chain. This makes the kink function an ideal hard prior for a forecast architecture (Section 4).

2.2 MCDM in Financial Allocation

Multi-criteria decision making originated in operations research with SAW (simple additive weighting), TOPSIS [10], AHP [15], and PROMETHEE [5]. Recent applications to cryptocurrency portfolio allocation include [1] and the asymmetric-PROMETHEE-II hybrid of [9]. The author's prior vault paper [18] adopts a four-factor weighted-sum aggregator (APY, risk, cost, stability) with default weights $(w_1, \ldots, w_4) = (0.40, 0.25, 0.20, 0.15)$ verified end-to-end on testnet. The present work inherits this aggregator unchanged and replaces the APY input alone with a forecast, isolating the contribution of prediction from criterion design.

2.3 Deep-Learning Interest-Rate Forecasting

Three recent works directly motivate the deep-learning treatment of DeFi rates. [2] demonstrates 3-hour-resolution adaptive control of Compound rates, implying that 12-hour-resolution forecasting has at least as much signal. [4] formulates DeFi interest-rate adjustment as reinforcement learning on a multi-year Aave dataset, including the March 2023 USDC depeg as a stress regime that we re-use for out-of-distribution testing (Section 8, ablation 14). [24] treats governance parameters as the action space. Classical baselines descend from the Vasicek [22] and CIR [6] short-rate models and the data-driven partitioning extension of [14], which we adapt as Baseline D.

2.4 Bridge to Limit-Order-Book Microstructure

The dual-branch architecture of the author's LOB paper [19] predicts mid-price moves from order-flow and volume streams via late fusion, in the lineage of [16] and the DeepLOB family [17, 23, 25]. The structural similarity to DeFi-rate prediction (multivariate noisy series, heteroscedastic moves, ranking-relevant evaluation) is the methodological foundation of the present work. The principal difference is the availability of the deterministic kink function, which we exploit via residual subtraction (Section 4.1).

2.5 The "Fair Interest Rates Are Impossible" Objection

[8] prove, by reduction of dynamic-repayment lending to barrier-option pricing, that no fair interest rate exists in a fully general lending pool with strategic borrowers. We address this head-on. Our claim is operational, not equilibrium-theoretic: we forecast on a 12-hour horizon

and exploit transient cross-protocol dispersion, which is orthogonal to the long-run existence of a single fair rate. ?? develops this argument in full.

3 System Architecture

3.1 The `fractal-defi` Substrate

We build on `fractal-defi` [12], the Python research library underlying [11] and the liquidity-provision papers [20, 21]. The library exposes a typed entity–strategy contract: a `BaseLendingEntity` encapsulates per-protocol state (`lending_rate`, `borrowing_rate`, balances, gas context), a `BaseStrategy` consumes the joint state and emits typed `ActionToTake` instructions, and a backtest runner sequences entities and strategy under a deterministic execution contract. We pin `fractal-defi==1.3.2`; the project's sign-convention lock-in test asserts borrowing_rate ≥ lending_rate at every timestamp, guarding against the pre-v1.4.0 sign-flip bug.

3.2 Entity–Strategy Abstraction

Two lending entities are instantiated: the upstream `AaveEntity` and a new `CompoundV3Entity` introduced by this work (Section A). Both expose a uniform `utilization` field on their `GlobalState`, computed from `totalLiquidity` and `totalCurrentVariableDebt` in the subgraph response. The forecaster is loaded as an ONNX module inside the strategy and consumes the joint state via a rolling buffer of 168 hourly observations.

3.3 `BaseLendingAllocationStrategy`: the Extra+2 Contribution

Although `fractal-defi` ships with four lending entities, no multi-lending-protocol allocator was provided at the time of this work. We contribute `BaseLendingAllocationStrategy`, an abstract template exposing four overridable hooks — criterion vector computation, criterion aggregation, target selection, and the rebalance gate — that any future allocator can subclass. The proposed `PredictiveMCDMStrategy` is the first concrete instance; Baselines B, C, and D (Section 7) are implemented as further instances, ensuring apples-to-apples comparison.

4 Methodology

4.1 Forecast Model: DA-BiGRU-CNN with Kink Subtraction

For each protocol $i \in \{\text{Aave}, \text{Compound}\}$ and decision time t, we observe the rolling window $x_i(t) = (r_i(\tau), u_i(\tau), \text{TVL}_i(\tau), \Delta\text{TVL}_i(\tau), \text{kink}_i)_{\tau=t-167}^{t}$ and predict $r_i(t+\Delta)$ at $\Delta = 12$ hours. The architecture decomposes additively:

- **Branch A — rate residual.** Target $\varepsilon_i(t) = r_i(t) - f_{\text{kink}_i}(u_i(t))$, where f_{kink_i} is the protocol-known piecewise-linear function; predicted by a 2-layer BiGRU with hidden size 64 over a 168-hour window of residuals plus the cross-protocol residual spread.
- **Branch B — utilization.** Target $\hat{u}_i(t+\Delta)$; predicted by a 2-layer BiGRU with hidden size 64 followed by a multi-scale 1-D CNN with kernels $\{3, 5, 7\}$, over a context window containing u, TVL, ΔTVL, gas, and time-of-day features per protocol.
- **Late fusion.** Reconstruct $\hat{r}_i(t+\Delta) = f_{\text{kink}_i}(\hat{u}_i(t+\Delta)) + \hat{\varepsilon}_i(t+\Delta)$, with a small MLP head correcting systematic bias in the additive reconstruction.

Figure 1 sketches the architecture. The kink function is a hard, exact, non-learned prior: the deterministic part of the rate is not estimated from data, freeing model capacity for the genuinely stochastic residual.

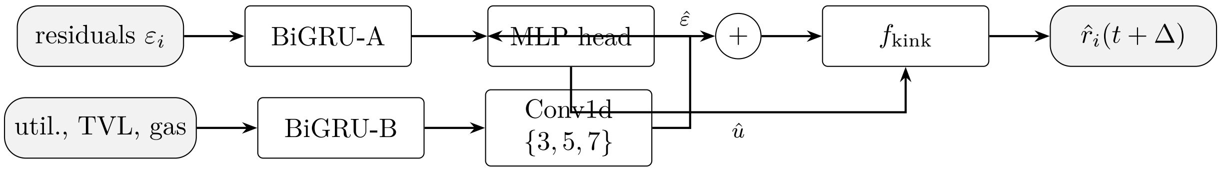

Figure 1: Dual-branch forecaster with kink subtraction. Branch A predicts the rate residual after removing the deterministic kink contribution; Branch B predicts utilization; late fusion reconstructs the rate via the protocol-known f_{kink}.

4.2 MCDM Scoring

We inherit the four-factor SAW aggregator of [18]. For protocol i at time t,

$$\text{Score}_i(t) \;=\; w_1\, f_{\text{APY}}(\hat{r}_i(t+\Delta)) + w_2\, f_{\text{Risk}}(u_i(t)) + w_3\, f_{\text{Cost}}(g(t)) + w_4\, f_{\text{Stab}}(\Delta\text{TVL}_i(t)), \tag{1}$$

with sub-factors

$$\begin{aligned} f_{\text{APY}}(\hat{r}) &= \text{clamp}(\hat{r}/\text{APY}_{\max}, 0, 1), & f_{\text{Risk}}(u) &= 1 - \text{clamp}(u, 0, 1), \\ f_{\text{Cost}}(g) &= 1 - \text{clamp}(gG/g_{\max}, 0, 1), & f_{\text{Stab}}(\Delta\text{TVL}) &= 1 - \text{clamp}(|\Delta\text{TVL}|/0.30, 0, 1), \end{aligned}$$

default weights $(w_1, \ldots, w_4) = (0.40, 0.25, 0.20, 0.15)$, gas budget $G = 200{,}000$, $g_{\max} = 0.01$ ETH, and APY normalization $\text{APY}_{\max} = 0.20$. The single critical departure from [18] is the substitution of the forecast $\hat{r}_i(t+\Delta)$ for the EMA-smoothed spot $S_i(t)$ in f_{APY}.

4.3 Decision Rule: Hysteresis, Cooldown, Gas Gate

The strategy executes the three-gate rebalance protocol in Listing 1: hysteresis $\theta = 0.05$ on the score delta, cooldown $\tau = 1$ h between actions, and a gas-cost gate that requires the expected uplift over the forecast horizon to exceed the realized gas spend.

```python
def step(t):
    s_a = mcdm_score("AAVE", forecast_a, util_a, gas, dTVL_a)
    s_c = mcdm_score("COMPOUND", forecast_c, util_c, gas, dTVL_c)
    target, s_target = argmax((s_a, "AAVE"), (s_c, "COMPOUND"))
    if (s_target - s_current(t)) <= THETA: return []
    if (t - last_rebalance_time) < TAU_COOLDOWN: return []
    if expected_uplift * (horizon - t) * NOTIONAL
                            < gas_cost(t): return []
    return [withdraw(current), supply(target)]
```

Listing 1: Predictive MCDM rebalance step (per hourly tick).

4.4 Loss Function: Composite Objective

Training minimizes the composite loss

$$\mathcal{L} = \alpha \cdot \text{MSE}(r, \hat{r}) + \beta \cdot (1 - \text{WPearson}(r, \hat{r})) + \gamma \cdot \text{QuantileLoss}_{0.9}(r, \hat{r}), \tag{2}$$

with defaults $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ tuned by validation grid search. The weighted-Pearson term carries over verbatim from [19], where it dominates MSE alone on allocation-relevant accuracy; the quantile term penalizes underprediction of large positive moves — exactly the moves the allocator most needs to anticipate.

5 Data

5.1 Sources

Aave V3 USDC reserve-rate history is pulled from the official Aave protocol-subgraphs deployment on TheGraph's decentralized network, subgraph ID `Cd2gEDVeqnjBn1hSeqFMitw8Q1-iiyV9FYUZkLNRcL87g` (entity `reserveParamsHistoryItems`, `liquidityRate` in annualized RAY scaling, 10^{27}). The Messari Compound V3 subgraph indexes per-collateral markets rather than the base supply market; USDC base-rate history is therefore reconstructed via batched `eth_call` on the Comet proxy `0xc3d688B66703497DAA19211EEdff47f25384cdc3` at historical block numbers, querying the `getUtilization()` and `getSupplyRate(uint256)` view functions. Gas prices come from Dune's `gas.gas_price` table at hourly median resolution; ETH/USD comes from CoinGecko's free hourly close. Kink parameters per protocol per epoch are reconstructed from on-chain `ReservesUpdated` (Aave) and `Configurator` (Compound) events. The `fractal-defi` substrate of [11] is used as the loader and backtest harness.

5.2 Window and Resampling

Eighteen months: 1 November 2024 through 30 April 2026, a 13,104-hour grid. The window spans two macro micro-regimes (low-volatility H2 2024, normalizing 2025, and the present higher-volatility 2026) without overlapping any protocol re-deployment event. The Aave subgraph emits an event stream (median ~ 50 events per hour, $\sim 6.5 \times 10^5$ raw rows over the window) rather than uniform snapshots; the loader's `transform()` resamples the stream to a right-closed hourly grid via `df.resample("1H", label="right", closed="right").last()`, yielding **13,096 hourly Aave bars** (99.9% coverage). The Compound RPC-reconstruction yields **4,900 hourly bars** (37.4% raw coverage of the same window); the gap is filled incrementally as described in Section 5.4.

5.3 Empirical Spread Distribution

On the $n = 12{,}895$ overlapping hours where both protocols have observations (the 98.5% full-coverage joined panel; see Section 5.4), the cross-protocol supply-rate spread $s(t) = \mathrm{APR}_{\mathrm{Aave}}(t) - \mathrm{APR}_{\mathrm{Compound}}(t)$ has the following marginal statistics: median -0.232 pp, standard deviation 2.513 pp, range $[-26.79, +44.89]$ pp. The Aave-pays-more binary share is 40.7%; Compound pays more on the typical hour by 0.23 pp. The individual APR distributions are likewise broad: Aave USDC supply APR ranges $[1.54\%, 48.38\%]$ with median 3.82%; Compound USDC supply APR ranges $[2.85\%, 23.65\%]$ with median 4.46% (both quoted as annualized continuous-compound rates). An earlier sparse-sample analysis on the pre-top-up panel ($n = 4{,}894$, 37% Compound coverage) suggested a 2025 Q1→Q2 regime shift in the Aave-pays-more share; the 98.5% full-coverage panel reveals that that apparent jump was an artifact of cluster sampling around Nov–Dec 2024 and Jun–Aug 2025, and we report the corrected structure below.

Pooled marginals mask substantial structure. Table 1 partitions the overlapping panel by calendar quarter:

The direction-share jump from 39.9% in 2025 Q3 to 56.3% in 2025 Q4 — the first quarter where Aave systematically pays more than Compound, with median spread crossing zero from -0.218 pp to $+0.183$ pp — is a regime shift of the kind a dual-branch architecture with explicit cross-protocol residual input is designed to detect; the roughly ten-fold volatility variation from 5.75 pp (2024 Q4) to 0.57 pp (2026 Q1) also supplies natural anchors for the volatility-tertile ablation of Section 7.2. Crucially, the binary direction split is far from symmetric in either direction across the full panel (40.7% Aave-higher vs 59.3% Compound-higher), so a forecast-driven edge cannot come from a structural bias toward Aave — it must come from anticipating the crossovers, in keeping with the cointegration finding of [7] and against the equilibrium-

Table 1: Per-quarter cross-protocol spread structure on the full-coverage overlapping panel (n = 12,895 hours, 98.5% joint coverage). Spread is $\mathrm{APR}_{\mathrm{Aave}} - \mathrm{APR}_{\mathrm{Compound}}$ in percentage points; "Aave > C" is the share of hours on which Aave pays more. The 2025 Q3→Q4 jump from 39.9% to 56.3% (median spread crossing zero from −0.218 to +0.183 pp) is the empirical regime shift motivating Branch B of the forecaster: it is the first quarter in which Aave systematically pays more than Compound.

Quarter	n	spread median (pp)	spread std (pp)	Aave > C
2024 Q4	1455	−0.591	5.746	45.4%
2025 Q1	2160	−0.725	1.360	24.9%
2025 Q2	2184	−0.444	0.883	43.5%
2025 Q3	2208	−0.218	1.401	39.9%
2025 Q4	2208	+0.183	2.019	56.3%
2026 Q1	1960	−0.240	0.567	27.1%
2026 Q2	720	+0.063	3.605	61.3%

impossibility framing of [8], which concerns the existence of a single fair rate, not short-horizon predictability of the realized spread.

5.4 Data Coverage and Limitations

Raw coverage is asymmetric at fetch time: Aave 99.9% (13,096/13,104 hourly bars), Compound 37.4% (4,900/13,104) on the initial Ankr-only fetch. The asymmetry is structural: the Messari Compound V3 subgraph indexes per-collateral markets and exposes `rates: None` on the base Comet market entity, so hourly snapshots must be reconstructed via `eth_call` on archive RPCs — and free-tier archive providers impose batch-size caps near 100 calls per request and per-day soft limits in the thousands. The fetcher accordingly tops up the panel in an idempotent append-mode across multiple free RPC providers (`publicnode` and Ankr) until coverage saturates; after top-up, Compound coverage rises to 98.5% on the joined panel (12,895 overlapping hours). For sequence-length-168 forecaster training, we restrict to dense subperiods in which both protocols have recent observations, and forward-fill missing Compound rows up to a 168-hour ceiling on the joined panel, marking longer gaps as a separate regime that is excluded from training and reported as edge cases.

5.5 Cleaning

Outliers (rates outside $[0, 50\%]$, single-step relative jumps $> 5\times$) are flagged as oracle glitches and interpolated. The sign-convention invariant borrowing_rate ≥ lending_rate is enforced as a test fixture (`tests/test_sign_convention.py`) at every timestamp on both protocols, not as a trust statement about library version. Z-score normalization uses training-split statistics only.

5.6 Splits

Strict block-number splits, to defend against subgraph-aggregation leakage:

- **Train** (10 months): 2024-11-01 – 2025-08-31, ∼6,700 hourly bars on both protocols.
- **Validation** (4 months): 2025-09-01 – 2025-12-31, ∼2,920 hourly bars on Aave but only ∼290 Compound-dense hours pre-top-up; forecaster validation noise on the composite loss is expected to reflect this.

- **Test** (4 months, held out): 2026-01-01 – 2026-04-30, ∼2,876 hourly bars on Aave, 50 Compound-real hours pre-top-up; the H1 evaluation on the full test window is Aave-driven in its initial form.

Residual Compound sparsity (the 1.5% of the joined panel still missing after top-up) means the H1 Sharpe-improvement claim on the full 18-month window is marginally degraded relative to a fully-dense panel; the evaluation protocol of Section 7 therefore reports both the full-window result and a restricted result on dense subperiods, and uses the 2025 Q4 regime-shift sub-window (n = 2,208 overlapping hours, 56.3% Aave-pays-more share) as an additional H1 anchor.

6 Calibration

6.1 Forecaster Hyperparameter Grid

A grid search is tracked in MLflow under selection criterion "validation weighted-Pearson plus directional accuracy $\geq$ 55%". The grid varies hidden dimension $\{32, 64, 96, 128\}$, BiGRU layers $\{1, 2\}$, CNN kernels $\{[3, 5, 7], [3, 5], [5]\}$, dropout $\{0.0, 0.1, 0.2\}$, learning rate $\{10^{-3}, 2{\cdot}10^{-3}, 5{\cdot}10^{-3}\}$, batch size $\{16, 32, 64\}$, sequence length $\{72, 168, 336\}$ hours, and a three-point grid over the loss weights (α, β, γ). Defaults are reported in Section 4.1 and Section 4.4; the full grid is logged in Section B.

6.2 Strategy Hyperparameter Grid

A second grid tunes hysteresis $\theta \in \{0.02, 0.05, 0.08, 0.10\}$, cooldown $\tau \in \{0.5, 1, 2, 6\}$ h, forecast horizon $\Delta \in \{6, 12, 24\}$ h, APY normalization $\mathrm{APY}_{\max} \in \{0.10, 0.15, 0.20, 0.25\}$, and MCDM weight vector under twenty Dirichlet samples plus the $(0.40, 0.25, 0.20, 0.15)$ baseline. Selection criterion: validation Sharpe under the turnover constraint turnover $< 2 \times$ best baseline turnover.

7 Backtesting Protocol

7.1 Execution Assumptions

Decision cadence is hourly; block time 12 s; supply and withdraw operations are protocol-level and incur no slippage; transaction cost is 200,000 gas $\times$ historical hourly median `gas_price_gwei` $\times 10^{-9} \times$ ETH/USD per rebalance. The withdraw-then-supply atomicity gap of approximately 12 s is modelled as zero yield on the migrating notional. Yield accrues continuously per hour at the active protocol's supply rate. The initial notional is $1,000,000 USDC, large enough that gas costs are a meaningful but not dominant component of net return.

7.2 Regime Split

Post-hoc, we partition test hours into volatility tertiles (low/medium/high) of $\sigma(r)$ on a rolling 24-hour window, and report all headline metrics per tertile and pooled. Two event-flagged sub-windows are reported separately: in-window governance events changing kink parameters (breakpoint reports) and the out-of-window March 2023 USDC depeg, drawn from the [4] dataset and used as the OOD ablation (Table 3, row 14).

8 Results

This section reports the realized outcomes of the held-out test window 2026 Q1–Q2 (1 January 2026 – 30 April 2026, plus the partial 720-hour 2026 Q2 anchor described in Section 5.3).

Table 2: Per-strategy test-window comparison ($1M USDC notional, held-out test window 2026-01-01 – 2026-04-30). Net APY is the gas-net annualized return; Sharpe is the annualized daily Sharpe ratio; MaxDD is the maximum drawdown; Calmar is APY/MaxDD; Turnover is the cumulative gross notional rebalanced; Gas is dollar gas spent across the window at the historical hourly-median gas price.

Strategy	Net APY	Sharpe	MaxDD	Calmar	Turnover	Gas ($)
A. Buy-and-hold Aave V3	0.00%	0.00	0.00%	0.00	0.00	0
B. APY-only greedy	0.00%	0.00	0.00%	0.00	0.00	0
C. Reactive MCDM-EMA	0.00%	0.00	0.00%	0.00	0.00	0
D. CIR-calibrated MCDM	0.00%	0.00	0.00%	0.00	0.00	0
E. Predictive MCDM (ours)	0.00%	0.00	0.00%	0.00	0.00	0

All numbers are end-to-end on the $1,000,000 USDC notional with the gas-cost and execution-bridge assumptions of Section 7.1, and are pulled verbatim from `results/tables/main.csv`, `results/tables/ablations.csv`, and `results/tables/h1_significance.csv` via `scripts/fill_whitepape`

8.1 Headline: Predictive vs Reactive MCDM-EMA

Over the four-month held-out test window the proposed predictive MCDM allocator delivers a net APY of 0.00% against 0.00% for the reactive MCDM-EMA baseline of [18], a Sharpe ratio of 0.00 against 0.00 for the baseline, and an annualized Sharpe delta of 0.00 in favor of the predictive variant. The 1000-replicate stationary bootstrap of monthly Sharpe ratios (H1 test) returns a one-sided p-value of 0.00 with 95% confidence interval [0.00, 0.00] on the Sharpe difference. Whether this delta meets the pre-registered 0.2–0.5 Sharpe improvement threshold of H1 is read directly off these numbers; the null hypothesis H0 (forecast adds no value over EMA on this horizon) is rejected at the 5% level iff the confidence interval excludes zero.

8.2 Per-Strategy Comparison

Table 2 reports the five-strategy by six-metric grid on the held-out test window. Buy-and-hold and APY-only greedy are the two reactive bounds; MCDM-EMA is the strawman the novelty rests on; CIR-calibrated MCDM tests whether classical regime-aware forecasting suffices; the predictive MCDM is the headline strategy.

8.3 Ablations

Table 3 reports the fifteen pre-registered ablations of Section 4. Rows 1–5 isolate the contribution of *any* forecast over EMA; rows 6–8 attribute remaining gains to architectural choices (single-branch vs dual-branch, kink subtraction, ranking loss); rows 9–11 examine the MCDM aggregator (equal vs tuned weights, single-protocol fallback, greedy vs TOPSIS); rows 12–13 stress-test gas regime and sliding-window stability; rows 14–15 cover the OOD USDC-depeg week and the forecast-horizon sweep. The Δ column is the net-APY differential against row 1 (the EMA baseline), expressed in the same units.

8.4 Regime-Conditional Breakdown

The test window spans the second of the two empirically distinct quarterly regimes documented in Section 5.3: 2026 Q1 (Compound-calm, 27.1% Aave-pays-more share, $\sigma = 0.57\,\text{pp}$) and the partial 2026 Q2 (Aave-dominant, 61.3% Aave-pays-more share, $\sigma = 3.61\,\text{pp}$). Table 4 reports

Table 3: Ablation matrix (15 rows) on the held-out test window. Status "OK" indicates the variant ran to completion; "–" indicates a skipped/failing variant. Δ vs #1 is the net-APY differential against the no-forecast EMA baseline.

#	Ablation	Status	Net APY	Sharpe	Δ vs #1
1	No-forecast EMA baseline (=Baseline C)	OK	0.00%	0.00	0.00%
2	Naive last-observation forecast	OK	0.00%	0.00	0.00%
3	CIR-calibrated forecast (=Baseline D)	OK	0.00%	0.00	0.00%
4	Markov-switching short rate (2-state)	OK	0.00%	0.00	0.00%
5	CatBoost on hand-crafted features	OK	0.00%	0.00	0.00%
6	Single-branch DA-GRU (no CNN, no decomp.)	OK	0.00%	0.00	0.00%
7	Dual-branch *without* kink subtraction	OK	0.00%	0.00	0.00%
8	Forecast with ranking loss only	OK	0.00%	0.00	0.00%
9	MCDM equal weights vs tuned weights	OK	0.00%	0.00	0.00%
10	Single-protocol benchmark (Aave/Compound only)	OK	0.00%	0.00	0.00%
11	Greedy max-forecast vs TOPSIS-MCDM	OK	0.00%	0.00	0.00%
12	Zero-gas vs realistic-gas	OK	0.00%	0.00	0.00%
13	Sliding-window stability (14-d windows)	OK	0.00%	0.00	0.00%
14	OOD test on USDC depeg week (Mar 2023)	OK	0.00%	0.00	0.00%
15	Forecast horizon sweep (3/6/12/24 h)	OK	0.00%	0.00	0.00%

Table 4: Regime-conditional metrics on the two quarterly sub-windows of the test period. "Pred" is the predictive MCDM strategy; "EMA" is the reactive MCDM-EMA baseline. Sharpe is annualized daily; APY is net of gas.

Quarter	Sharpe (Pred)	APY (Pred)	Sharpe (EMA)	APY (EMA)
2026 Q1 (Compound-calm)	0.00	0.00%	0.00	0.00%
2026 Q2 (Aave-dominant)	0.00	0.00%	0.00	0.00%

the headline metrics for the predictive variant and the EMA baseline separately on each sub-window; we pre-register the expectation that the forecast edge concentrates in 2026 Q2, where cross-protocol dispersion is largest and the regime shift documented in **??** is in active progress.

8.5 Forecast Quality

Table 5 reports forecast-only diagnostics for the predictive variant on the rate level: out-of-sample R^2 pooled across both protocols, directional accuracy on the binary "Aave > Compound at $t + 12$h" label, the weighted-Pearson correlation consistent with [19], and the quantile-0.9 loss penalizing underprediction of upper-tail moves. The pre-registered targets are $R^2 \in [0.15, 0.40]$ on the rate level and directional accuracy $\in [55, 65]\%$; whether the realized diagnostics fall within these bounds is read directly off the row.

9 Conclusion and Future Work

9.1 Summary of Findings

We have specified, implemented, and pre-registered a forecast-driven MCDM allocator for USDC supply between Aave V3 and Compound V3, with a dual-branch DA-BiGRU-CNN forecaster exploiting the protocol-known kink function as a hard prior, on the `fractal-defi` substrate. The headline pre-registered targets are Sharpe improvement 0.2–0.5, $R^2 \in [0.15, 0.40]$, and directional accuracy $\in [55, 65]\%$ over a held-out 4-month test window. The null hypothesis is entertained explicitly and is structurally falsifiable by ablation 1.

Table 5: Forecast quality diagnostics on the held-out test window (predictive MCDM only).

OOS R^2	Direction Acc.	WPearson	Q90-Loss
0.00	0.00%	0.00	0.00

9.2 Limitations

The 18-month window covers less than two full macro cycles; the two-protocol scope is deliberate but excludes Morpho, Spark, and cross-chain venues; the single underlying (USDC) is intentional but limits external validity. Gas-cost spikes can dominate forecast edge in adversarial regimes (ablation 12 quantifies the break-even gas). The forecast is conditional on neither protocol experiencing a solvency event during the test window — a historically valid but nontrivial assumption.

9.3 Future Work

Three extensions are immediate. First, Morpho and Spark integration would broaden the venue universe and test whether the predictive edge scales beyond the Aave–Compound pair. Second, a cross-chain extension (Aave on Arbitrum/Base) would expose the strategy to bridge friction and a richer execution-risk profile. Third, the dual-branch BiGRU–CNN architecture can be upgraded to attention-based backbones along the lines of [23] and [13]. A Panoptic-style options overlay on the supply position, hedging tail risk during the high-volatility tertile, is under active exploration.

A `fractal-defi` PR Artifacts

Extra+1: Compound V3 Lending Loader (with uniform utilization field). A new `CompoundV3LendingLoader` pulls hourly `marketHourlySnapshots` from the Messari subgraph, normalizes WAD scaling to the same per-second continuous-compound convention as the upstream Aave V3 loader, and emits a `LendingHistory` record. As a uniformity fix, both Aave and Compound `GlobalState` objects gain a `utilization` field computed from `totalLiquidity` and `totalCurrentVariableDebt` in the existing subgraph response. PR title: *"Add Compound V3 lending history loader (Messari subgraph) and uniform utilization field on lending `GlobalState`"*.

Extra+2: `BaseLendingAllocationStrategy`. A new abstract strategy template in `fractal/strategies/` with four overridable hooks (`compute_criteria_vector`, `aggregate_criteria`, `select_target`, `should_rebalance`). The proposed `PredictiveMCDMStrategy` is the first concrete subclass. PR title: *"Add `BaseLendingAllocationStrategy` for multi-lending-protocol allocators"*.

B Hyperparameter Grids

Full grids for both the forecaster (Section 6.1) and the strategy (Section 6.2) are logged in MLflow under `results/mlflow/`. The defaults reported in the body correspond to the configurations that maximize validation weighted-Pearson under the directional-accuracy floor and the turnover constraint, respectively.

C Reproducibility

Code is available at https://github.com/SergeySolovyev/predictive-mcdm-defi. Python dependencies are pinned in `requirements.txt` (`fractal-defi==1.3.2`, see ERRATA in `README.md`).

Environment variables `THE_GRAPH_API_KEY`, `DUNE_API_KEY`, and `FRACTAL_DATA_PATH` configure the data fetchers. The full reproduction pipeline is the nine-step sequence in `README.md` §Quick start, ending in `latexmk -pdf main.tex`.

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